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ME 1060: DYNAMICS

Lecture 1: Introduction to Dynamics

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SLIIT

Acknowledgement: These lecture slides are originally prepared by Dr Hansani Weeratunge(SLIIT)

What is this subject about?

This course teaches the foundation principles necessary to understand and predict the motion of objects under the influence of various forces, which is critical in designing and analysing mechanical systems.

Assessment Activities

Mid Semester (1 hour) – 20%
Assignments and Quizzes – 15%
Laboratories – 15%
End of semester examination (2 hours) – 50%

Important Guidelines

Please note the following important guidelines:

- All communication will be through email only.
- Always include "ME1060 2026:" at the beginning of the subject line in your email.
- Submit all lab reports through CourseWeb only. If you face a valid issue, you may email your report to the relevant instructor and CC me in the email.

- A penalty of 5% per day will be applied for delayed submissions.

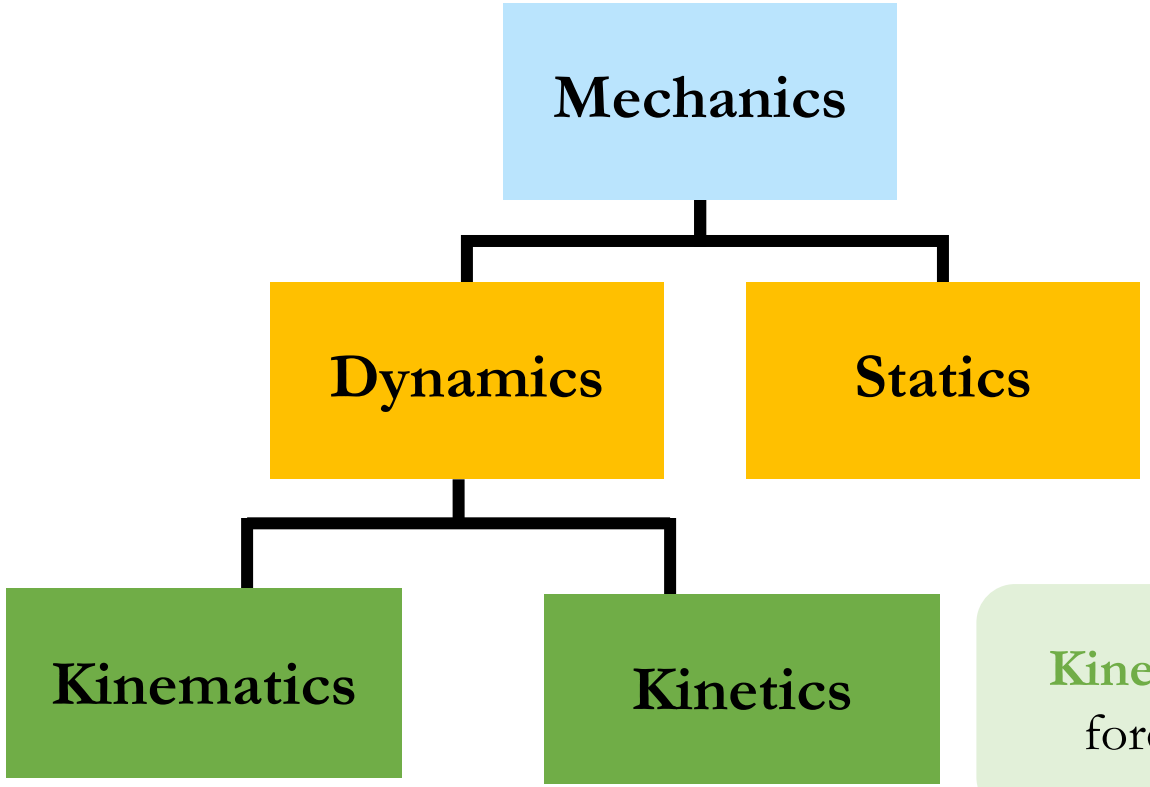
- **All assignments must be individual work. Plagiarism will not be tolerated under any circumstances.**

Overview

- **Introduction to dynamics**
  - Introduction, History, Applications
  - Basic concepts (space, time, mass, particle, rigid body)
  - Kinematics of a particle
    - Rectilinear motion
    - Curvilinear motion
- **Supporting Reading Material**
  - Hibbeler, R.C. 2016, Engineering Mechanics Dynamics, 14th Edition, Pearson. (ISBN: 978-0-133-91538-9)
  - Meriam, J.L., and Kraige, L.G. 2008, Engineering Mechanics, Vol. 2: Dynamics, SI Unit, 6th Edition, John Wiley & Sons. (ISBN: 978-0-471-78703-7)

Introduction

Mechanics: a branch of the physical sciences that is concerned with the state of rest or motion of bodies subjected to the action of forces.

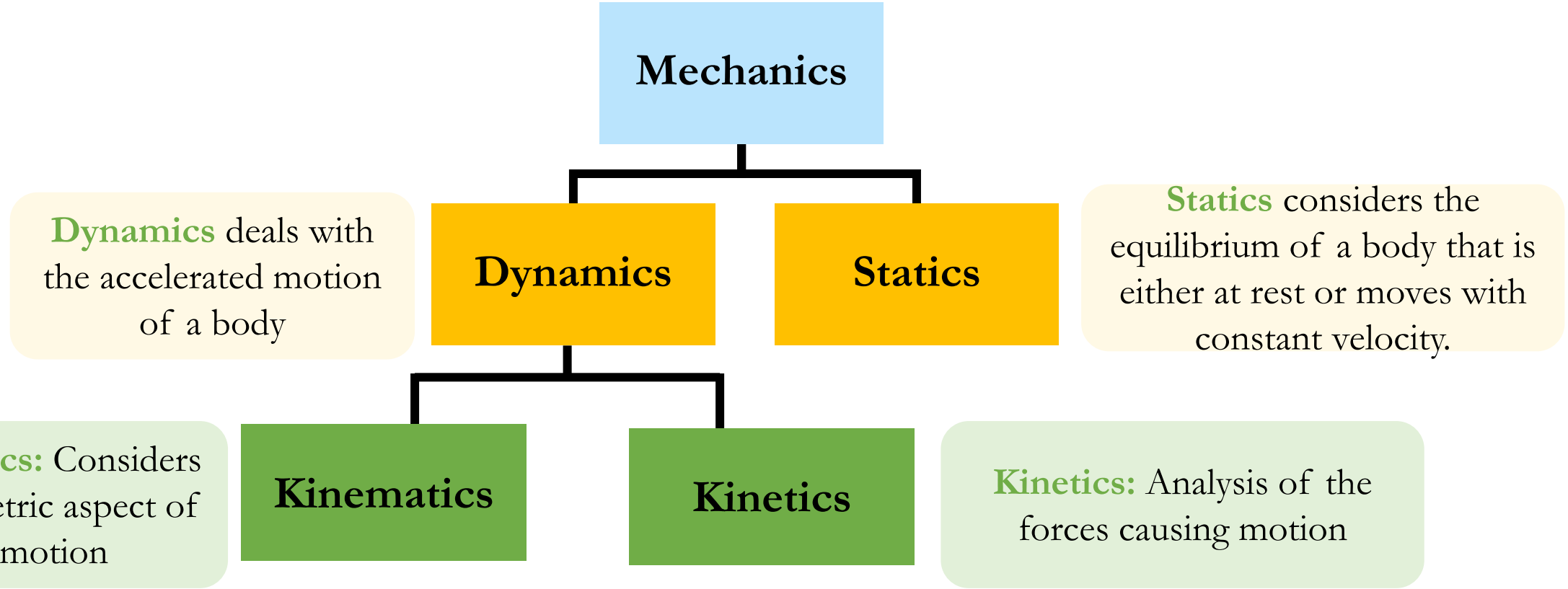


Kinematics: Considers the geometric aspect of the motion

Kinetics: Analysis of the forces causing motion

Introduction

Mechanics: a branch of the physical sciences that is concerned with the state of rest or motion of bodies subjected to the action of forces.



Dynamics has two distinct parts;

Kinematics

The study of motion without reference to the forces which cause the motion and treats only the geometric aspects of motion.

Kinetics

The analysis of the forces which cause the motion.

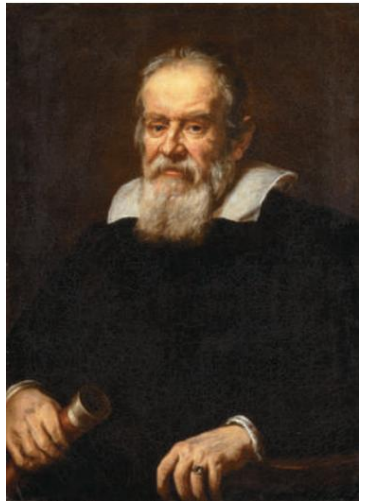
Students of engineering will find that a thorough comprehension of dynamics will provide one of their most useful and powerful tools for analysis in engineering.

History

Brief History of Dynamics

"Historically, dynamics is a relatively recent subject compared to statics. The beginning of a rational understanding of dynamics is credited to Galileo (1564-1642), who made careful observations concerning bodies in free fall, motion on an inclined plane, and motion of the pendulum.

He was largely responsible for bringing a scientific approach to the investigation of physical problems. Further significant developments in dynamics awaited the invention of the pendulum clock by Huygens in 1657."



Newton (1642-1727), guided by Galileo's work, was able to make an accurate formulation of the laws of motion and, hence, to place dynamics on a sound basis. Newton's famous work was published in the first edition of his Principia, which is generally recognized as one of the greatest of all recorded contributions to knowledge.

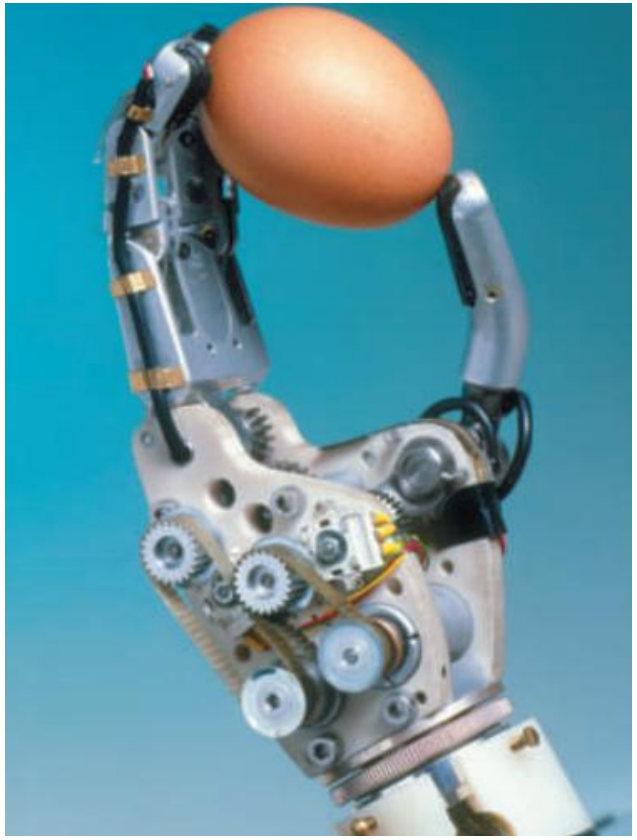
In addition to stating the laws governing the motion of a particle, Newton was the first to correctly formulate the law of universal gravitation.

After Newton, important contributions to mechanics were made by Euler, D'Alembert, Lagrange, Laplace, Poinsot, Coriolis, Einstein, and others.

Applications of Dynamics

Only since machines and structures have operated with high speeds and appreciable accelerations, has it been necessary to make calculations based on the principles of *dynamics* rather than on principles of *statics*.

The rapid technological developments of the present day require increasing application of the principles of mechanics, particularly dynamics.



These principles are basic to

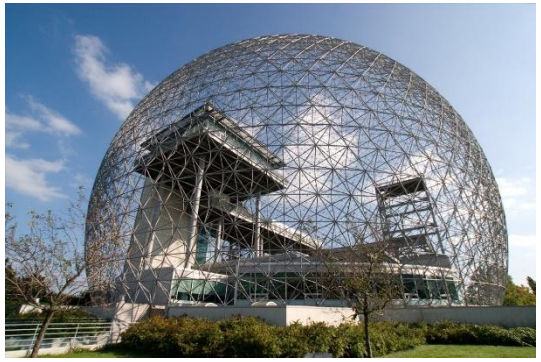
- the analysis and design of moving structures
- fixed structures subject to shock loads
- robotic devices
- automatic control systems
- rockets, missiles, and spacecraft
- ground and air transportation vehicles
- electron ballistics of electrical devices,

and to machinery of all types such as

turbines, pumps, internal combustion (reciprocating and rotary) engines, hoists, machine tools, etc.

Static Structures







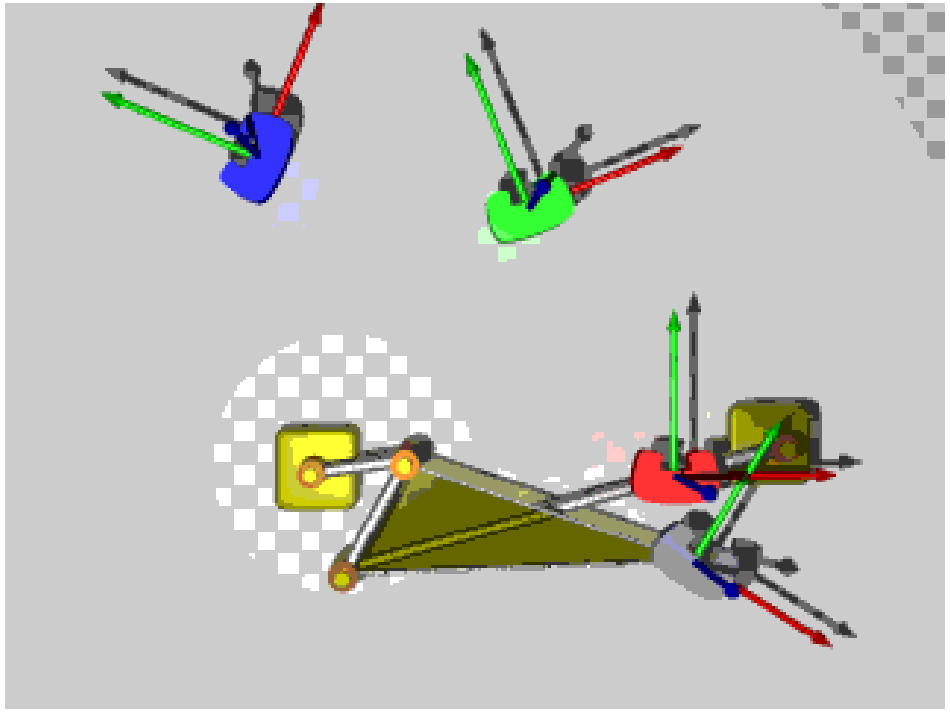




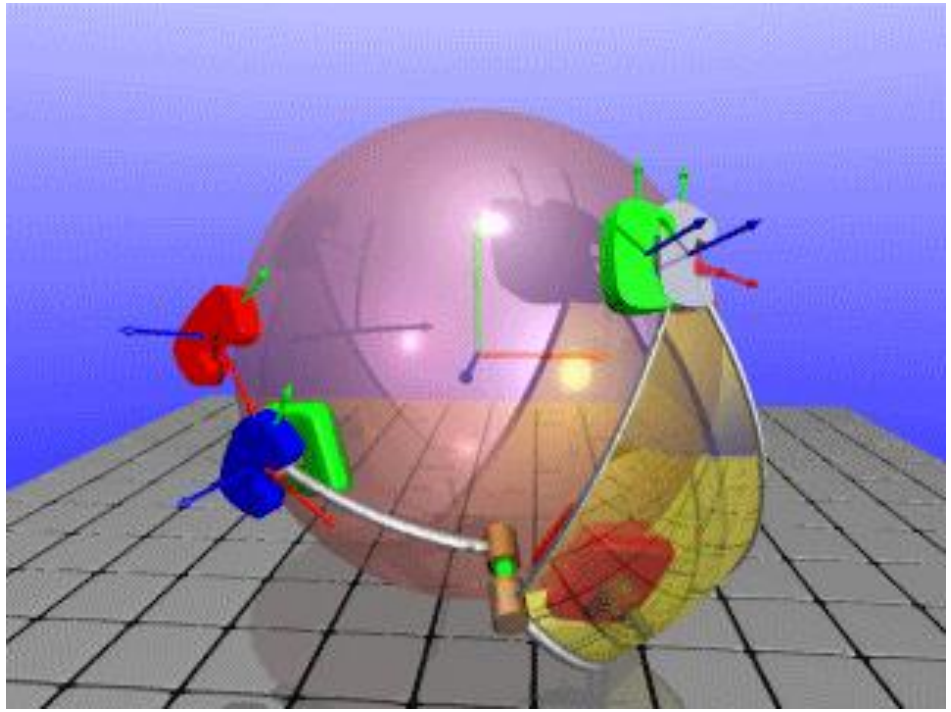




Dynamic Structures

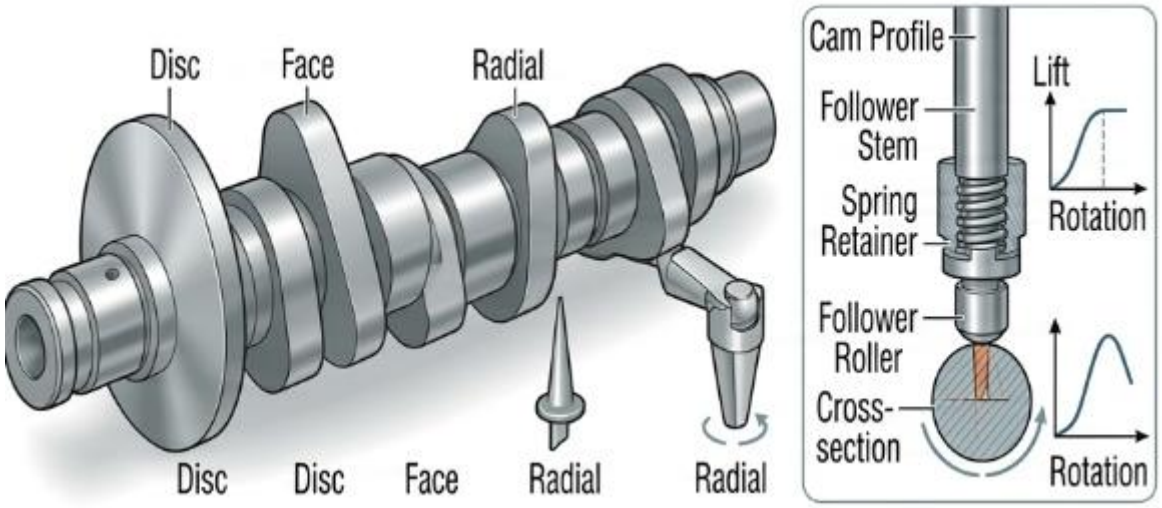


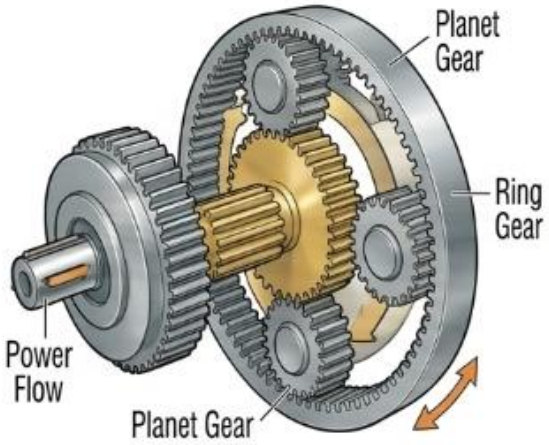
4 bar link mechanism



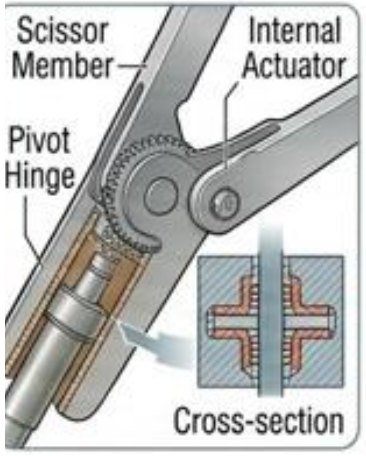
4 bar chain (3D spherical)

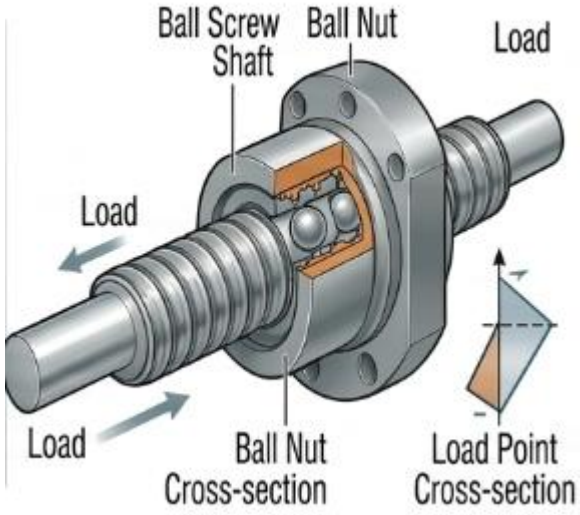
Dynamic Structures











Basic Concepts

Space – Geometric region occupied by bodies.

Position in space is determined relative to some geometric reference system by means of linear and angular measurements. The basic frame of reference for the laws of Newtonian mechanics is the **primary inertial system** or **astronomical frame of reference**, which is an imaginary set of rectangular axes assumed to have no translation or rotation in space.

Measurements made with respect to this reference are said to be absolute, and this reference system may be considered **"fixed"** in space





Basic Concepts

Time – is a measure of the succession of events and is considered an absolute quantity in Newtonian mechanics.

Mass - is the quantitative measure of the inertia or resistance to change in motion of a body. Mass may also be considered as the quantity of matter in a body as well as the property which gives rise to gravitational attraction.

Force - is the vector action of one body on another.

Basic Concepts

Particle – is a body of negligible dimensions. When the dimensions of a body are irrelevant to the description of its motion or the action of forces on it, the body may be treated as a particle. An airplane, for example, may be treated as a particle for the description of its flight path.

Rigid body – is a body whose changes in shape are negligible compared with the overall dimensions of the body or with the changes in position of the body as a whole.

Kinematics of a particle

A particle has a mass but negligible size and shape.

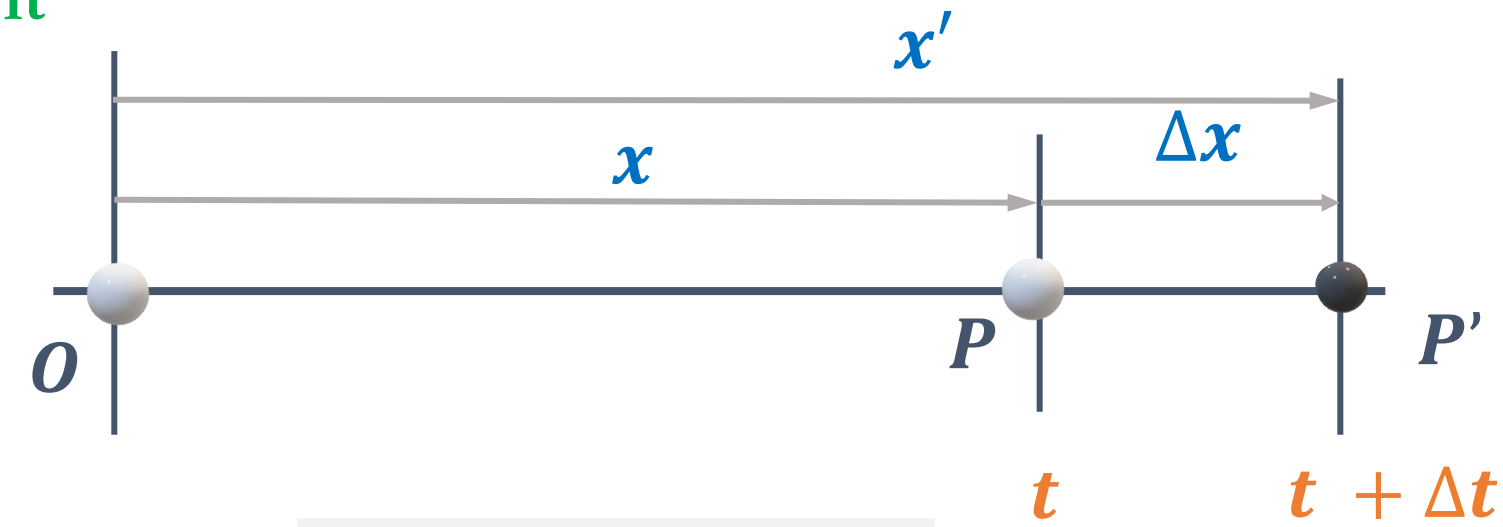
In most problems in dynamics, one is interested in bodies of finite size, such as **aircraft, projectiles, or vehicles.**

Such objects may be considered as particles, **provided motion of the body is characterized by motion of its mass center and any rotation of the body is neglected.**

Rectilinear motion

A particle moving along a straight line is said to be in **rectilinear motion**

Displacement



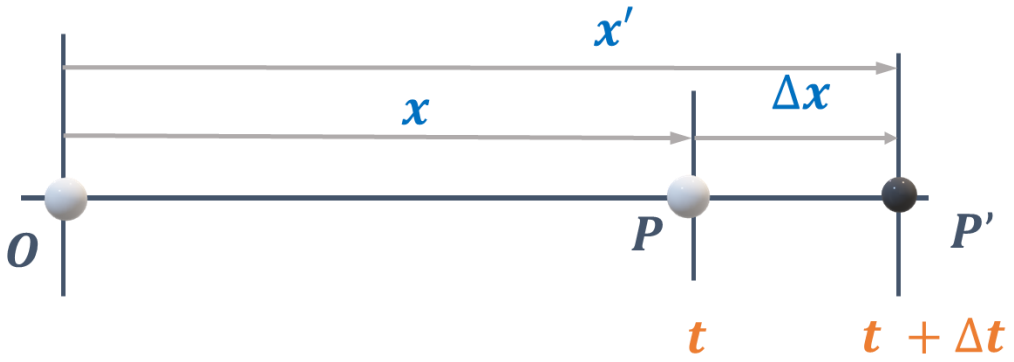
$$\Delta v = \frac{\Delta x}{\Delta t}$$

The displacement of a particle is also a **vector quantity**, and it should be distinguished from the distance the particle travels.

Velocity

If the particle moves through a displacement Δx during the time interval Δt,

Average velocity of the particle over the time interval Δt ;



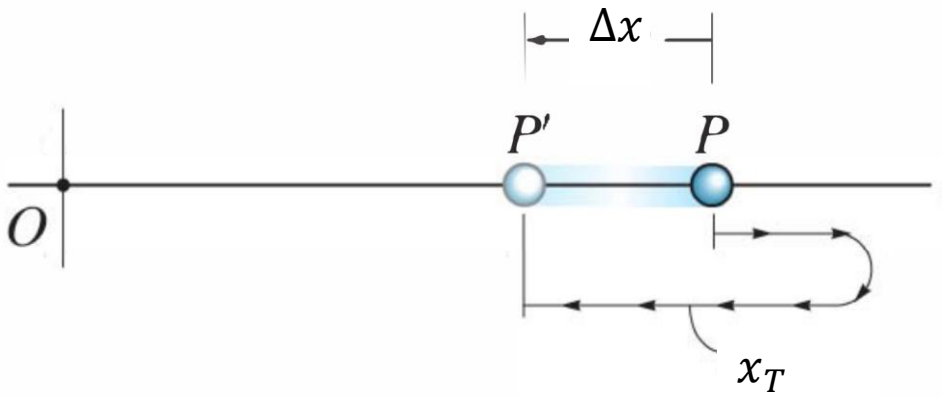
$$v = \frac{\Delta x}{\Delta t}$$

Since Δt or dt is always positive, the sign used to define the direction of the velocity is the same as that of Δs or ds.

The magnitude of the velocity is known as the speed, and it is generally expressed in units of m/s or ft/s.

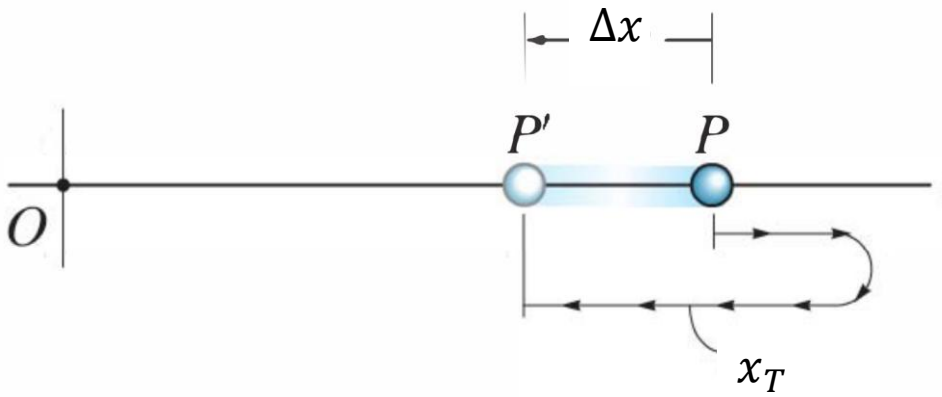
Velocity

Particle in the following figure travels along the path of length x_T in time Δt. Find the particles average speed and average velocity.



Velocity

Particle in the following figure travels along the path of length x_T in time Δt. Find the particles average speed and average velocity.



Average Speed $(v_{sp})_{avg} = \frac{x_T}{\Delta t}$

Average Velocity $v_{avg} = \frac{-\Delta x}{\Delta t}$

If we take smaller values of Δt, the magnitude of Δs becomes smaller and smaller.

Consequently,

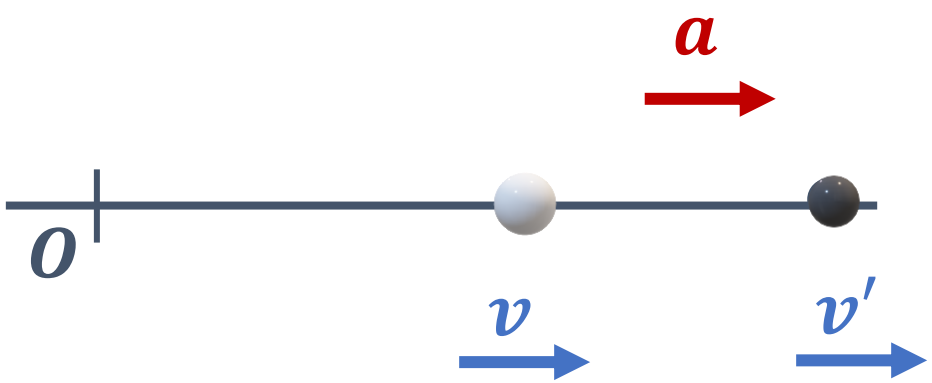
Instantaneous velocity of the particle at time t ;

$$v = \lim_{\Delta t \to 0} \left(\frac{\Delta x}{\Delta t} \right) = \frac{dx}{dt} = \dot{x}$$

Acceleration

Similarly, if Δv is the increment of velocity over the time interval Δt.

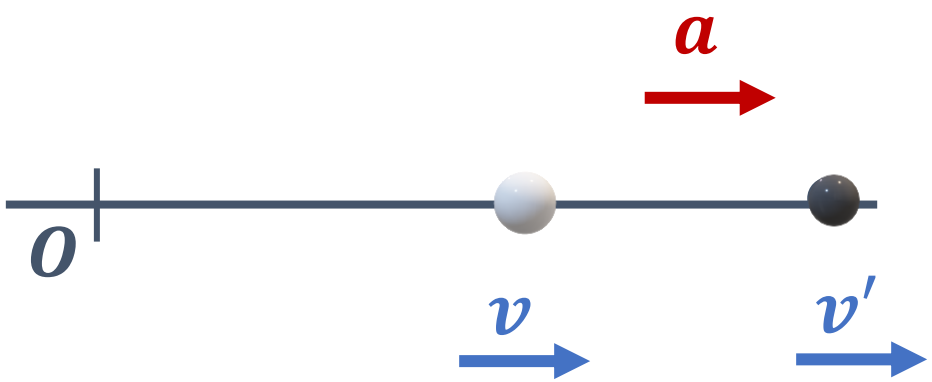
Instantaneous acceleration of the particle at time t ;



Acceleration

Similarly, if Δv is the increment of velocity over the time interval Δt.

Instantaneous acceleration of the particle at time t ;



$$\boldsymbol{a} = \lim_{\Delta t \to 0} \left(\frac{\Delta \boldsymbol{v}}{\Delta t} \right) = \frac{d\boldsymbol{v}}{dt} = \frac{d^2\boldsymbol{x}}{dt^2} = \ddot{\boldsymbol{x}}$$

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \frac{d\boldsymbol{x}}{d\boldsymbol{x}}\frac{d\boldsymbol{v}}{dt} = \frac{d\boldsymbol{x}}{dt}\frac{d\boldsymbol{v}}{d\boldsymbol{x}} = \boldsymbol{v}\frac{d\boldsymbol{v}}{d\boldsymbol{x}}$$

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt}$$

$$\boldsymbol{a}\, d\boldsymbol{x} = \boldsymbol{v}\, d\boldsymbol{v}$$

Constant Acceleration

Constant Acceleration $a = a_c$

When the acceleration is a constant where $a = a_c$, each of the three kinematic equations $a_c = dv/dt$, $v = ds/dt$, and $a_c\, ds = v\, dv$ can be integrated to obtain formulas that relate $a_c, v, s,$ and t.

Velocity as a function of time:

Integrate $a_c = dv/dt$, assuming that initially $v = v_0$ when $t = 0$.

$$\int_{v_0}^{v} dv = \int_{0}^{t} a_c dt$$

$$v = v_0 + a_c t$$

Position as a function of time

Integrate $v = ds/dt = v_0 + a_c t$, assuming that initially $s = s_0$ when $t = 0$.

$$\int_{s_0}^{s} ds = \int_{0}^{t} (v_0 + a_c t) dt$$

$$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$$

Velocity as a function of position

integrate $v\,dv = a_c\,ds$, assuming that initially $v = v_0$ at $s = s_0$.

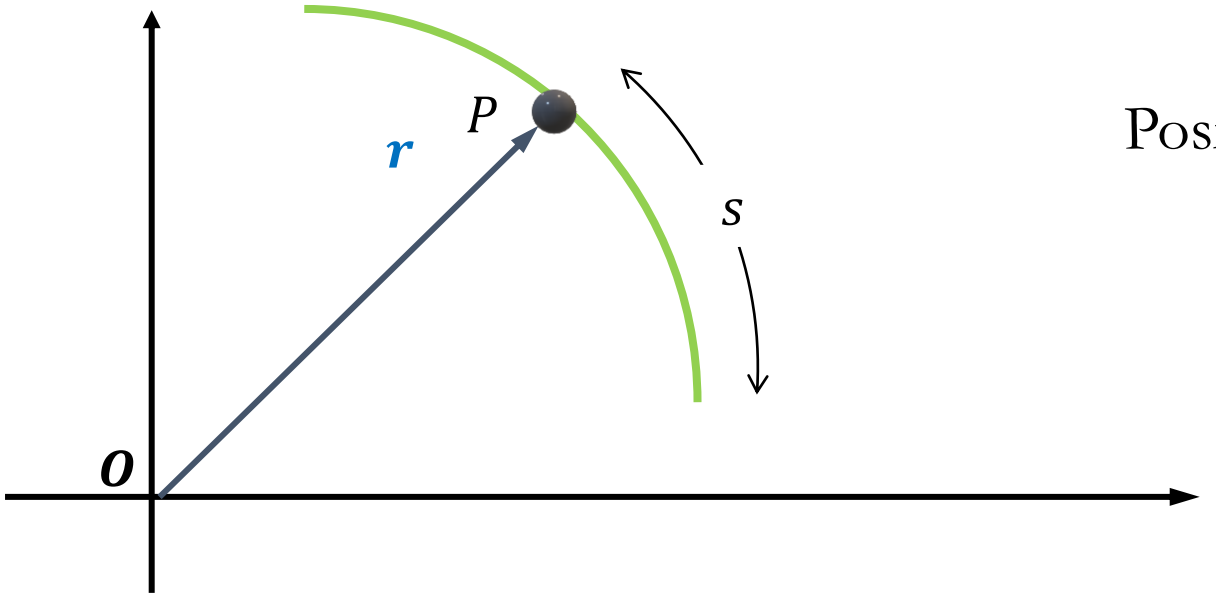
$$\int_{v_0}^{v} v dv = \int_{s_0}^{s} a_c ds$$

$$v^2 = v_0^2 + 2a_c(s - s_0)$$

Curvilinear motion

A particle is said to be in curvilinear motion when it moves along a curve other than a straight line.

Position

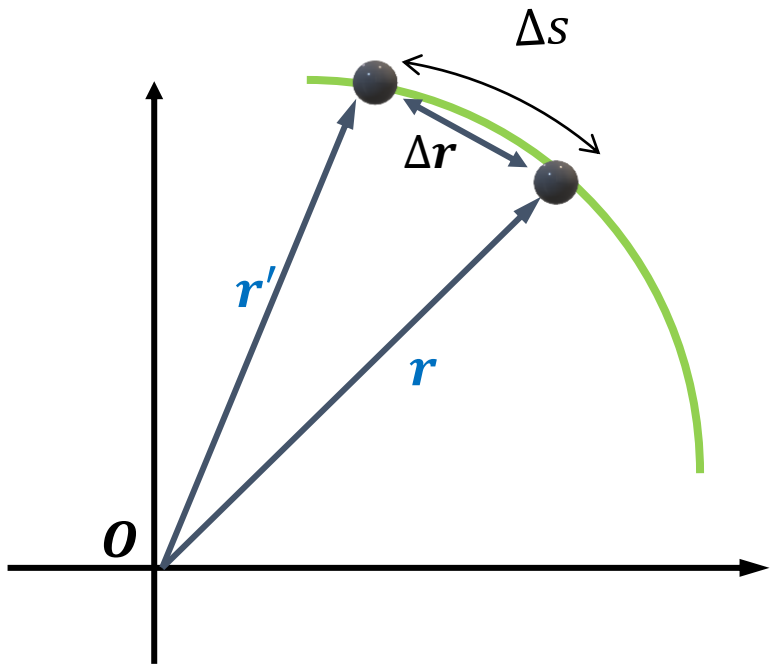


Position Vector $\boldsymbol{r} = \boldsymbol{r}(\boldsymbol{t})$

Vector $\boldsymbol{r}$ is the position vector of the particle.

Displacement

Suppose that during a small time interval Δt the particle moves a distance Δs along the curve.



The new position $\boldsymbol{r' = r + \Delta r}$
Therefore,

The displacement $\boldsymbol{\Delta r = r' - r}$

Average Velocity

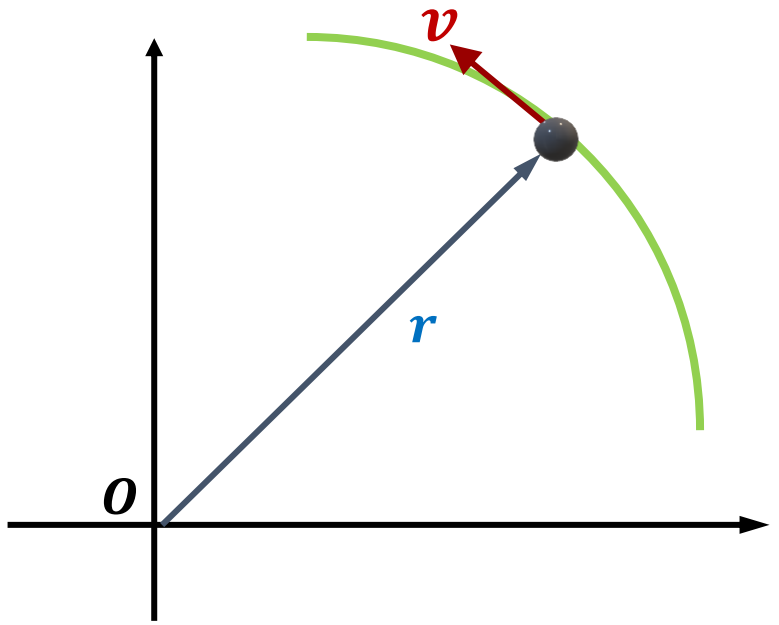
During Δt, the average velocity is

$$\boldsymbol{v_{avg}} = \frac{\boldsymbol{\Delta r}}{\Delta t}$$

Instantaneous Velocity

When $\Delta t \to 0$, the instantaneous velocity is

$$\boldsymbol{v} = \lim_{\Delta t \to 0} \left(\frac{\Delta \boldsymbol{r}}{\Delta t} \right) = \frac{d\boldsymbol{r}}{dt} = \dot{\boldsymbol{r}}$$

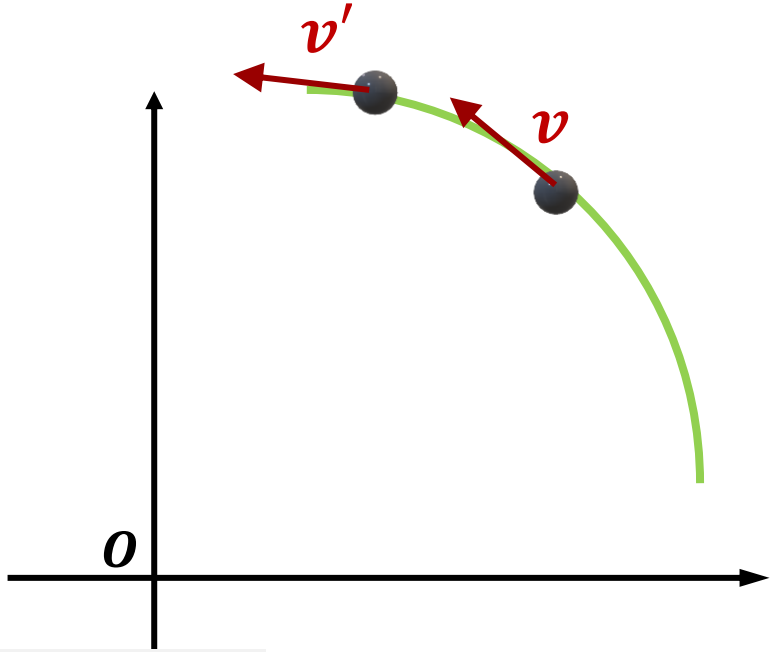


Since $d\boldsymbol{r}$ will be tangent to the curve, the direction of $\boldsymbol{v}$ is also tangent to the curve.

The magnitude of $\boldsymbol{v}$ i.e. speed v

$$\text{Speed } v = |\boldsymbol{v}| = \frac{ds}{dt} = \dot{s}$$

Acceleration



If the particle has a velocity v at time t and a velocity $v' = v + \Delta v$ at $t + \Delta t$, then

Average acceleration $\boldsymbol{a_{avg}} = \dfrac{\Delta \boldsymbol{v}}{\Delta t}$

Iinstantaneous acceleration $\boldsymbol{a} = \lim_{\Delta t \to 0} \left(\dfrac{\Delta \boldsymbol{v}}{\Delta t}\right) = \dfrac{d\boldsymbol{v}}{dt} = \dfrac{d^2\boldsymbol{r}}{dt^2}$

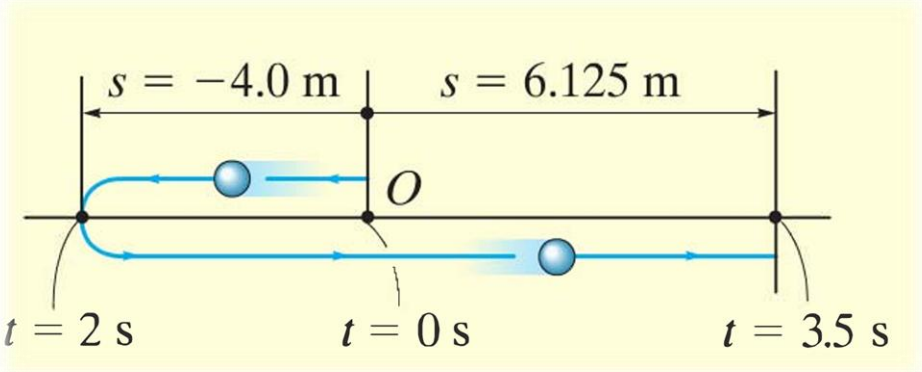
The acceleration a, then, includes the effects of both the change in magnitude of v and the change of direction of v. It is apparent, in general, that the direction of the acceleration of a particle in curvilinear motion is neither tangent to the path nor normal to the path.

Example 1

A particle moves along a horizontal path with a velocity of $v = (3t^2 - 6t)\ m/s$, where t is the time in seconds. If it is initially located at the origin o, determine the distance travelled in $3.5\ s$, and the particle's average velocity and average speed during the time interval.

Example 1

A particle moves along a horizontal path with a velocity of $v = (3t^2 - 6t)\ m/s$, where t is the time in seconds. If it is initially located at the origin o, determine the distance travelled in $3.5\ s$, and the particle's average velocity and average speed during the time interval.



Summary

Important Points

- Dynamics is concerned with bodies that have accelerated motion.
- Kinematics is a study of the geometry of the motion.
- Kinetics is a study of the forces that cause the motion.
- Rectilinear kinematics refers to straight-line motion.
- Speed refers to the magnitude of velocity.
- Average speed is the total distance traveled divided by the total time. This is different from the average velocity, which is the displacement divided by the time.
- A particle that is slowing down is decelerating.
- A particle can have an acceleration and yet have zero velocity.
- The relationship $a\,ds = v\,dv$ is derived from $a = dv/dt$ and $v = ds/dt$, by eliminating dt.